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ITA-0329: MOBILE COMPUTING

CIA-1:

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4.)
Sol:

Given,

A cellular operator deploys a hexagonal network.

$= 1200 km^2$.

Cell Radius = 1.8 km

Available spectrum = 48 MHz

per full duplex channel = 200 KHz.

Reuse cluster sizes, N=7 & N=4

Reduction usable channels = 15%.

$= 0.15$

Minimize reuse ratio $\Rightarrow Q = 4.5$

No. of cells:

$$A_C = \frac{3\sqrt{3}}{2} R^2$$

$$= 2.598 (1.8)^2 \quad (\because \text{Cell Radius } (R) = 1.8)$$

$$= 8.5197 km^2$$

$$= \frac{1200 km^2}{8.5197 km^2} \approx 143 \text{ cells.}$$

∴ The Approximate No. of cells = 143.

Total Available channels:

$\Rightarrow$ Available Spectrum / per full Duplex channel

W.K.T, $1\ MHz = 10^3\ KHz$.

$$= \frac{24}{48} \times 10^3\ KHz$$

$$= 240$$

∴ Total Available channels = 240.

Effective channels

$\Rightarrow 240(1 - 0.15)$

⇒ Reduction due to interface & guard channel observations.

$= 240(0.85)$

$= 204$

∴ Effective channels = 204.

Observations:

Configuration	N=7	N=4.
Theoretic channels per cell	240	240
Effective channels per cell	204	204
Reuse Distance. $D = R\sqrt{3N}$	$D = 1.8\sqrt{3\times7} = 1.8\sqrt{21}$	$D = 1.8\sqrt{3\times4} = 1.8\sqrt{12}$
Reuse Ratio $= \sqrt{3N}$	$= \sqrt{3\times7} = \sqrt{21} = 4.58$	$= \sqrt{3\times4} = \sqrt{12} = 3.58$
No. Of Approximate cells	≈143 cells.	≈143 cells

Given, Minimize reuse ration $Q = 4.5$

$\Rightarrow R \geq Q$.

$N = 7$, $4.58 \geq 4.5$ ✓ (condition satisfied)

$N = 4$, $3.58 \geq 4.5$ ✗ (condition not satisfied).

Hence,

> $N = 7$, satisfies the minimum reuse ratio.

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(2) sol

Given,

Two public Land Mobile Networks (PLMNs)

During 40 minutes emergency operation.

operation-1 PLMN-1 — operation-2 PLMN-2

No. of cell boundaries, => 15

Speed => 90 km/h — 40 km/h

PLMN Handoff => 120 ms

NSS path switching => 80 ms.

Additional transition => 350 ms

Authentication = 250 ms.

=> Three Handoffs failed each requires additional = 400 ms for recovery.

potential data loss at 512 kBps.

1) Handoff Delay = $T_r = 180 * 80$

$= 200$ ms.

for 15 cell boundaries, $200 \times 15 = 3000$ ms

$\Rightarrow$ 3 seconds

Additional Recovery ~~Handoff~~ Delays

$\Rightarrow 3 \times 400 = 1200$ MS.

$T_h \Rightarrow 1.2$ S.

Roaming data Delays

$T_h = (350 * 250)/2 = 1.2$ seconds.

Interrupted Data Delays

$= 3.0 + 1.2 + 1.2$

$= 5.4$ seconds.

potential data Delays

$\Rightarrow 350 \times 5.0 \times 250$

$\Rightarrow 345$ bits.

limit check: 345 bits x 3S $\Rightarrow$ 5.4S Satisfied.

$\therefore$ The data covery limit check 345 bits

3S $<=>$ 5S is match the system & cover the limit.

(3) i) Given,

Base Transceiver stations = 12.

Sectors = 3

Transceivers = 4

Each TRX = 8 TDMA slots.

⇒ 2 timeslots per sector.

Each sector serves = 1200 subscribers/hour.

1.2 calls per hours. can holding 90 seconds.

Each channel Requires 16 kbs.

Single e1 link = 2048 Mbps.

⇒ Traffic channel per sector = 4 x 8 x 2

⇒ 64

Traffic channel per BTS ⇒ 30 x 3

= 90.

Traffic channel / BSC = 90 x 12

⇒ 1080.

Offered Traffic sector A $= \frac{1200 \times 1.2 \times 90}{3600}$ $\frac{12 \times 30}{36A}$

$= 36A$.

Channel vigualization (U) $= \frac{36}{36} \times 100$

= 100 %.

Required channel (80%) $= \frac{36 \times 0.8}{100}$ $\frac{36 \times 0.8}{288}$

$= 288$

Min TRX/sector $= 8 - 2$

$= 6$ TRXs/sector.

with 6TRX channel $\Rightarrow 6 \times 8 - 2$

$\Rightarrow 46$.

New utilization $= \frac{36}{48} \times 100 = 75\%$.. $(\leq 80\%)$.

Between Abis per Bits. $\Rightarrow 90 \times 16$.

$= 144$ Mbps.

Abis upGrade $= 46 \times 3 \times 16$.

$= 2.2$ Mbps.

∴ Increase TRX from 4 to 6 sector 46 channels &.

75% utilization After upGrade Abis 2.2 Mbps.

So, E1 is insufficient.

(ii) Given, one way $= 10$ MHz.

$N = 7$

frequency $= 0.4$ & 0.6

(4) Given,

Network transfers = 240 image files.

Each size = 18 MB.

24×18

$3 \quad 192$

$24 \quad 432$

$\Rightarrow 240 \times 18$

Total image file consumption = 4320 MB

Each providing a theoretical dataRate = 20 kbps.

Reduced capacity = 18%.

Effective throughput = 85%.

Transmission Rate increases = 12%.

lasting = 2.5 seconds.

UMTS time = 384 kbps.

GPRS meets the = 12 hour deadline.

Total Data = 240×18 = 4320 MB.

GPRS = $4 \times 20 = 80$ kbps.

Effective GPRS = $80 \times (1 - 0.18) \times 0.85$

= 55.76 kbps.

Total transmitted volume =

$4320 \times (1 + 0.12)$

= 4384. MB.

GPRS = 192.83h > 12h $\Rightarrow$ Not satisfied.

UMTS = 25h > 12h $\Rightarrow$ ~~Not~~ satisfied.

(5) Given, software update = 120 GB.

out of 2000 Mobile devices.

$= 2000 \times 120$

$= 240000$ GB.

GSM provides ⇒ 9.6 kbps per connection.

GPRS provides ⇒ 64 kbps Additional 10% .. (6.4 kbps)

5G provides = 25 Mbps per device.

4% protocol overhead.

40% of devices can transfer data simultaneously over 5G.

$\Rightarrow 2000 \times \frac{40}{100} = 800$ Mobile devices.

Assuming 1 GB = 1000 MB.

1 MB = 8 megabits.

Devices(5G) update..

800 × 25 Mb/s.

per minute,

25 × 60 Mb/min.

1500 Mb/Min

per hour,

1500 × 60 Mb/h

90000 Mb/h.

⇒ 1 GB = 1000 MB.

per hour = 90 GB.

∴ Therefore,

Total time taking 5G Devices to update

= 1 hour 20 minutes.

$\frac{1.5 \times 32}{18000}$

18 Gn.

$\frac{1.5 \times 24}{60}$

$\frac{1.5 \times 20}{300}$.